

Hackathon Problem:

1. Data Pipelines

Build data pipelines and analyze the system for IoT-Based (Simulated) Smart Energy Consumption and Safety Monitor

- a. Simulate a system of 100s of IOT devices (simulate device output, not actual devices) to minimize energy usage and safety by utilizing various sensors in a large factory/Manufacturing plant.
- b. Steps to Achieve:
 - i. Simulated Hardware: 100s of sensors will send current, temperature, and pressure data at various points. (Bots will simulate this)
 - ii. Data Collection: Stream real-time sensor data to a server using Kafka/AWS
 - iii. Analytics: process the data to detect high-consumption or critical patterns.
 - iv. Optimization Suggestions: Use ML to recommend device usage schedules.
 - v. Predict system failure.
 - vi. Control Layer: Remotely turn devices on/off via relays. Relays will be simulated.
 - vii. UI Dashboard: Build a web/mobile dashboard with charts (Grafana, React).
 - viii. Extra Challenge: Integrate with smart assistants (Alexa/Google Home).

2. Artificial Intelligence

AI-Powered Real-Time Sign Language Translator for multilingual meetings.

- a. Goal: Build a system that captures sign language via webcam and translates it into text and speech in real time.
- b. Steps to Achieve:
 - i. Data & Model: Use an open dataset like RWTH-PHOENIX-Weather or any other or create a small custom dataset for common signs.
 - ii. Computer Vision: Implement hand and body pose detection.
 - iii. Model Training: Train or use a pre-trained gesture recognition model.
 - iv. Translation Layer: Map recognized gestures to words/phrases.
 - v. Speech Output: Use a TTS engine.
 - vi. UI: Build a web or mobile interface for live translation.
 - vii. Extra Challenge: Add reverse translation — speech-to-sign animation using 3D avatars.

3. Devops

Kubernetes

Build a mini system that deploys a microservice on Kubernetes clusters.

- a. If a pod crashes, it should detect, restart, and notify via Slack/Discord.
- b. Add a chaos test (kill pods randomly), and the system must self-recover.
- c. Constraints: Should include monitoring (Prometheus/Grafana) + auto-restart logic.
- d. Analyze system performance and predict failures.

4. Computer Vision

Identify adversarial input for AI

Develop an AI-Powered Traffic Violation Detection from CCTV Feeds. Detect traffic violations (signal jumping, wrong-way driving) automatically. You can use sample traffic footage or a live camera feed. YOLOv8 or Detectron2 can be used for vehicle detection. Define traffic rules and send violation snapshots to an admin dashboard.

With this trained ML model, build a tool that:

- a. Test it on adversarial or biased inputs.
- b. Shows explainability (SHAP / LIME style visualizations).
- c. Generate an audit report with model drift, fairness score, and accuracy.

5. Security

Develop guardrails for AI models.

AI companions for intimacy are chatbots and virtual avatars that users interact with for emotional support, friendship, and romantic connection, often through advanced language models that learn user preferences and provide tailored, supportive conversations. These AI partners offer continuous availability and user-centric interaction, which can help alleviate loneliness and meet unmet emotional needs; however, they also present significant risks, including user addiction, the potential for harmful suggestions, and the danger of forming unhealthy attachments.

Develop a system having proper guardrails for:

- a. Input filtering (toxicity, profanity, topic restrictions)
- b. Output filters/validation (check whether generated content aligns with retrieved evidence, no hallucination)
- c. PII detection/scrubbing
- d. Access controls, role-based policies
- e. Strong logging and audit trails to monitor failures.